Project Design Phase-II

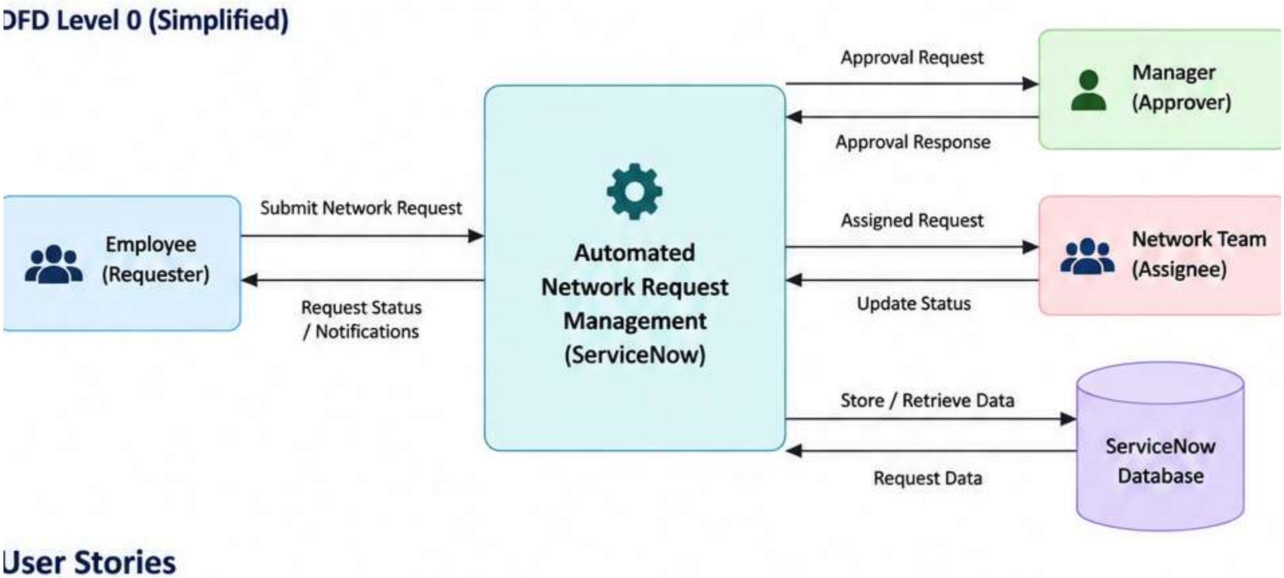
Data Flow Diagram & User Stories

Date	3 September 2026
Team ID	
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) represents how information moves through the system. For this project, it shows how a network request enters ServiceNow, moves through approval and assignment, is stored, and returns status information to the requester.

DFD Level 0 (Simplified)



Main flow: Employee submits request → ServiceNow processes → Manager approves → Network Team fulfills → status is updated → requester is notified.

User Stories

Use the below template to list the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Employee (Requester)	Submit Request	USN-1	As a user, I can submit a network request (VPN, Firewall, IP, etc.) through a Service Catalog form.	Request is created with a unique number.	High	Sprint-1
Employee (Requester)	Request Tracking	USN-2	As a user, I can track the current status of my network request.	Current status is visible on the portal.	High	Sprint-1
Manager (Approver)	Approval Management	USN-3	As a manager, I can approve or reject network requests.	Request moves to the correct next stage.	High	Sprint-1
Manager (Approver)	Approval Notifications	USN-4	As a manager, I receive notifications for pending approvals.	Email and in-platform notification is received.	Medium	Sprint-1
Network Team (Assignee)	Request Fulfillment	USN-5	As a network team member, I can view and work on requests assigned to my group.	Assigned requests appear in the correct queue.	High	Sprint-2
Network Team (Assignee)	Status Update	USN-6	As a network team member, I can update status and close completed requests.	Status is updated and requester is notified.	High	Sprint-2
System Admin	SLA & Escalation	USN-7	As an administrator, I can monitor SLA timers and configure escalation rules.	SLA status and escalation are triggered correctly.	High	Sprint-2
System Admin	Reporting	USN-8	As an administrator, I can view dashboards and reports for network requests.	Dashboard shows request volume, status and SLA trends.	Medium	Sprint-3

User Roles Covered

Employee (Requester) • Manager (Approver) • Network Team (Assignee) • System Administrator

Release Plan

Sprint-1: Request submission, tracking and approval | Sprint-2: Fulfillment, status updates, SLA and escalation | Sprint-3: Dashboards and reporting

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Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Network Request Submission	Service Catalog form; required network details; request category
FR-2	Request Validation & Priority	Validate mandatory fields; calculate impact/urgency; set priority
FR-3	Approval Workflow	Route approval to manager; approve/reject request; capture approval decision
FR-4	Automatic Assignment	Assign to correct network support group; create fulfillment task
FR-5	SLA & Escalation	Start SLA; track response/resolution time; escalate approaching or breached SLA
FR-6	Notifications & Tracking	Send submission/approval/status notifications; allow requester to track request
FR-7	Request Fulfillment & Closure	Update work notes/status; complete request; close after fulfillment
FR-8	Dashboard & Reporting	Show request volume, status, priority and SLA performance reports

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a simple Service Catalog form with clear fields, guidance and easy request tracking.
NFR-2	Security	Use role-based access so requesters, approvers, network teams and administrators see only authorized data and actions.
NFR-3	Reliability	Maintain consistent request, approval and assignment processing with proper error handling and audit history.

NFR-4	Performance	Request submission, status updates and workflow actions should complete within acceptable response time.
NFR-5	Availability	The request management process should be available during business operations with minimal service interruption.
NFR-6	Scalability	Support increasing users, requests, network teams and workflow volume without major redesign.

Project Scope: End-to-end automation of network request submission, approval, assignment, SLA management, notification, fulfillment and reporting in ServiceNow.

Project Design Phase-II
Technology Stack (Architecture & Stack)

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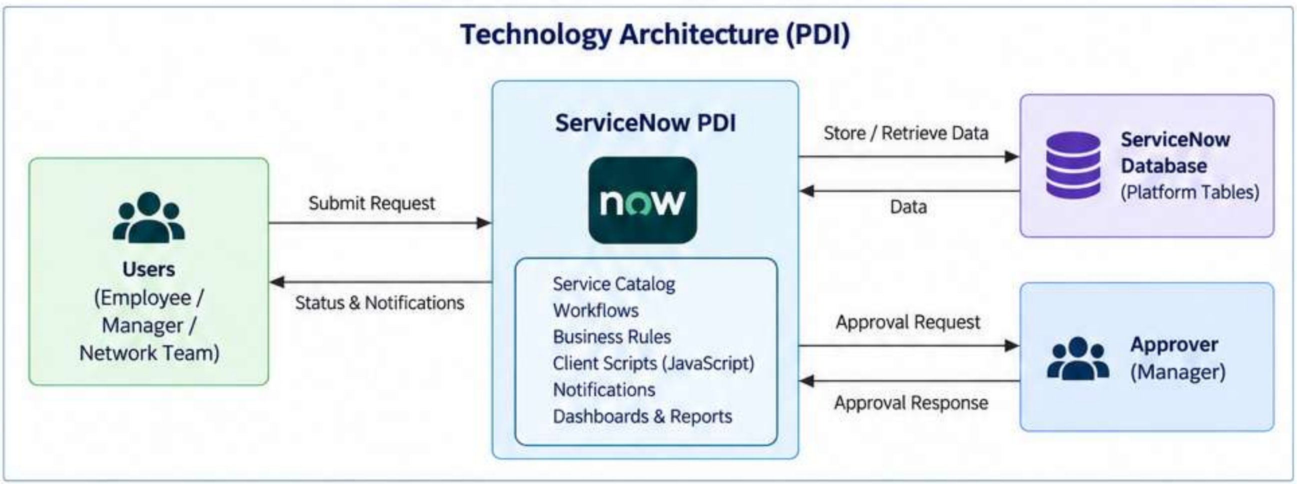
Technical Architecture:

The architecture uses a ServiceNow Personal Developer Instance (PDI) with simple ServiceNow configuration and basic JavaScript. The solution keeps the design lightweight and uses mainly out-of-box platform features.

Architecture Diagram (PDI)



Key flow: Employee → ServiceNow request form → validation/workflow → Manager approval → Network Team → database → notifications and reporting.

No machine learning model is required for the basic project scope.

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Technology Stack (Architecture & Stack)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology (PDI)
1	User Interface	Portal / Service Catalog used to submit network requests.	ServiceNow Service Catalog, Service Portal
2	Application Logic	Form validation, field logic and simple automation.	Basic JavaScript – Client Scripts, UI Policies
3	Workflow	Approval, assignment, notifications and request stages.	Flow Designer / Workflow
4	Database	Stores requests, approvals, tasks, SLA data and configuration.	ServiceNow Platform Tables
5	Notifications	Sends request, approval and status notifications.	ServiceNow Notification Engine
6	SLA Management	Tracks response and resolution targets.	ServiceNow SLA / Task SLA
7	Reporting	Displays request status, volume and SLA information.	ServiceNow Reports & Dashboards
8	External API (Optional)	Can connect to a network monitoring tool if required later.	REST API
9	Infrastructure	Development and testing environment for the project.	ServiceNow Personal Developer Instance (PDI)

Technology Summary

Primary technologies: ServiceNow PDI, Service Catalog, Flow Designer, Business Rules, Client Scripts, UI Policies, Notifications, SLA, Reports and basic JavaScript.

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Technology Stack (Architecture & Stack)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology (PDI)
1	Open-Source Frameworks	No additional open-source framework is required for the basic PDI implementation.	Basic JavaScript, HTML/CSS if customization is needed
2	Security Implementations	Only authorized users should create, approve, update or administer requests.	Roles, ACLs, User Criteria, ServiceNow security controls
3	Scalable Architecture	Uses the ServiceNow cloud platform and reusable workflows for growing request volume.	ServiceNow PDI / ServiceNow Platform
4	Availability	The solution runs on the ServiceNow platform rather than a self-managed local server.	ServiceNow Cloud Platform
5	Performance	Use simple scripts, mandatory fields and efficient queries; avoid unnecessary client/server processing.	ServiceNow best practices, GlideRecord where required

Implementation Scope

• Platform: ServiceNow Personal Developer Instance (PDI).
• Development: Basic JavaScript using Client Scripts and Business Rules.
• Automation: Flow Designer / Workflow for approval, assignment, SLA and notifications.
• Storage: ServiceNow platform tables; no separate database is required.
• Advanced AI/ML is outside the basic project scope.

References

https://c4model.com/ | https://www.servicenow.com/